effect of three forces - the pressure gradient force, the frictional force and the Coriolis force. In addition, the gravitational force acts downward.

68. When the pressure drops along the front, the warm air moves northwards and the cold air move towards, south setting in motion an anticlockwise cyclonic circulation. The cyclonic circulation leads to a well developed extra tropical cyclone, with a warm front and a cold front.

    The warm air glides over the cold air and a sequence of clouds appear over the sky ahead of the warm front and cause precipitation. The cold front approaches the warm air from behind and pushes the warm air up. As a result, cumulus clouds develop along the cold front.

69. Water vapour present in the air is known as humidity. It is expressed quantitatively in different ways. The actual amount of the water vapour present in the atmosphere is known as the *absolute humidity*. It is the weight of water vapour per unit volume of air and is expressed in terms of grams per cubic metre. The ability of the air to hold water vapour depends entirely on its temperature. The absolute humidity differs from place to place on the surface of the earth.

70. *Cirrus*

    Cirrus clouds are formed at high altitudes (800012,000m). They are thin and detatched clouds having a feathery appearance. They are always white in colour.

    Cumulus

    Cumulus clouds look like cotton wool. They are generally formed at a height of 4000-7000 m. They exist in patches and can be seen scattered here and there. They have a flat base.

    Stratus

    As their name implies, these are layered clouds covering large portions of the sky. These clouds are generally formed either due to loss of heat or the mixing of air masses with different temperatures.

    Nimbus

    Nimbus clouds are black or dark gray. They form at middle levels or very near to the surface of the earth. These are extremely